# UNIVERSITY OF OTAGO EXAMINATIONS 2022

## COMPUTER SCIENCE

### COSC343

### ARTIFICIAL INTELLIGENCE

**Semester Two**

**(TIME ALLOWED: THREE HOURS)**

This examination comprises 6 pages.

Candidates should answer questions as follows:

Candidates should answer **All** questions.

Questions are worth 12 marks and sub marks are shown thus: (2)

The total number of marks for this exam is 60.

The following material is provided:

Nil.

Use of calculators:

No restriction on the model of calculator that may be used, but no device with communication capability shall be accepted as a calculator. Calculators are subject to inspection by the examiners.

Candidates are permitted copies of:

Nil.

**TURN OVER**

## 1. State-space search

(a) What do *completeness* and *optimality* aspects of a state-space algorithm search relate to? (2)

(b) Are completeness and optimality guaranteed for A*? If so, are there any conditions that need to be met? If not, are there any aspects of A* that can make completeness and optimality more likely? (2)

(c) In general, does it make sense to talk about optimality in terms of the path through the state space when it comes to a Genetic Algorithm (GA)? Explain your answer. (2)

(d) Imagine that you are a consultant for a game company, which aims to develop a zero-sum two player board game. They want to provide an online version, where players can play against an AI. The AI must be settable to different levels of difficulty and capable of playing the optimal game in the hardest mode.

(i) What type of AI algorithm would be suitable here? Briefly describe how it works. (2)

(ii) How would your algorithm provide different levels of difficulty? Would anything specific to the game in question be needed for it to work? (2)

(iii) Given the mechanics of the game are not solidified yet, could you recommend anything about the nature of those mechanics that would make the optimal play possible and/or practically achievable by the AI agent? (2)

## 2. Probabilistic reasoning

(a) Explain what the equations $P(A|B,C) = \frac{P(B,C|A)P(A)}{P(B,C)}$ and $P(A,B|C) = P(A|C)P(B|C)$ are and how are they used in a Naïve Bayes Classifier? (4)

(b) Suppose you need to create a detector of the tone of individual sentences in an e-mail. The decision needs to be made whether a particular sentence is *formal*, *informal*, or *humorous*. Treating each sentence in an e-mail as a bag of words model, design a Naïve Bayes Classifier for this classification. What training data is needed for this, and what expression(s) need(s) to be evaluated for the decision on the tone of a sentence consisting of "$w_1\ w_2 \ldots w_N$", where $w_i$ is the $i^{\text{th}}$ word? (4)

(c) How should the case of encountering a word not found in any of the training examples be handled, so that the classifier can still make a decision based on other words in the sentence? (2)

(d) Why might very long sentences be problematic for a Naive Bayes Classifier implementation on a computer? What can be done to avoid a potential problem? (2)

## 3. Classification and regression

Consider the following four hypotheses:

$\hat{y}_1 = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0,$
$\hat{y}_2 = \sigma_s(w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0),$
$\hat{y}_3 = \sigma_h(w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0),$
$\hat{y}_4 = w_1 x_1 x_2 + w_2 x_2^2 + w_3 x_1^2 x_2 + w_4 x_3,$

where $\sigma_s(v) = \frac{1}{1+e^{-v}}$, $\sigma_h(v) = \begin{cases} 1 & v > 0 \\ 0 & v \leq 0 \end{cases}$, $x_i \in \mathcal{R}$, and $w_i \in \mathcal{R}$.

(a) Identify the model/hypothesis name that each is known by (use the index of $\hat{y}_i$ to clearly identify which name refers to which hypothesis)? (2)

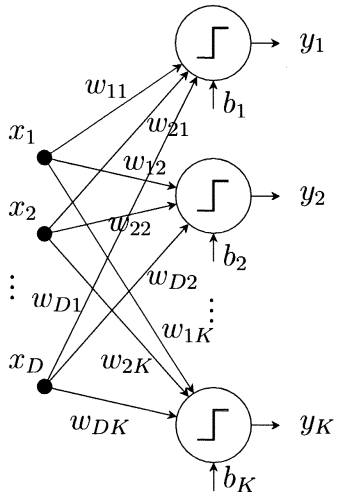
(b) For each of those hypotheses, identify whether it is best suited for classification, regression, or for both. What loss function would be appropriate to use for training in each case? (4)

(c) Which of the above can be trained to fit a supervised dataset with the linear Least Squares method and which one with Steepest Gradient Descent optimisation? (2)

(d) Assume you have a dataset with a data point $\mathbf{x} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$ with the corresponding target value $y = 0.75$. State which of the above hypotheses can be consistent on this dataset. For those that cannot be consistent, explain why. (4)

## 4. Neural networks

Consider the following diagram:




(a) What learning hypothesis does the above figure represent, and what type of general learning problem can it be used for? (2)

(b) What do the symbols $x_i$, $w_{ij}$, $b_j$, $y_j$, $K$, $D$ represent in the figure above? (3)

(c) What is the function $\sigma_h(v) = \begin{cases} 1 & v > 0 \\ 0 & v \leq 0 \end{cases}$ and how is it implied in the above diagram? (2)

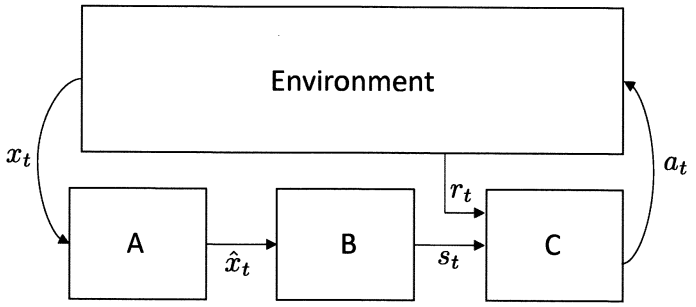
(d) Specify the configuration of the above model for digit recognition (from 0 to 9 inclusive) from a 16×16 grey-scale image. How would you label the images? (2)

(e) What is the fundamental limitation of this hypothesis? (1)

(f) What can be done to the architecture to turn it into a trainable universal function approximator? (2)

## 5. Unsupervised, generative and reinforcement learning

Consider the following diagram representing a reinforcement learning agent consisting of three modules, A, B, and C, receiving at time $t$ percept $x_t$ from the environment and producing corresponding action $a_t$:




(a) The percepts of the agent are sourced from a camera, which at time $t$ produces a 128× 64 grey-scale image $x_t$. The function of module A in the diagram above is to compress image $x_t$ into an 8-dimensional real valued vector $\hat{x}_t$. Assuming we want compression specific to the images encountered by the agent, what method do you propose for this? Explain how it would work and what (if any) training would be involved for this specific scenario. Mathematical details are not required, just a good description of the workings of the proposed method. (3)

(b) Module C contains a Q-table with 256 rows and three columns labelled $a = -1$, $a = 0$, and $a = 1$. The module receives two inputs: $s_t$, an integer value between 1 and 256 representing a state, and $r_t$, the reward signal for state $s_t$. What is the function of this table, what do its rows and columns represent, how is it used and how (if at all) is it trained? Mathematical details of the operation of the table and training are not required. (3)

(c) Can you think of a way for module B to convert an 8-dimensional vector $\hat{x}_t$ into $s_t$, an integer value between 1 and 256, so that the state feeding the Q-table meaningfully represents a category of a given $\hat{x}_t$? What method could you use to develop this categorisation? Explain how the method works and what training (if any) would be involved. Mathematical details are not required, just a good description of the workings of the proposed method (3)

(d) Suppose that, after all this effort, the agent is not performing well in the environment (on whatever task it is put to). Identify and explain possible effects of tweaks (one in each module) that could potentially help. (3)



**END**